



Preventable accidents in Indian coal mining: A socio-technical alignment approach to labour safety

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Abstract

With a significant position in the global minerals map, for India, the importance of the triple bottom line of people, planet, and profit in Indian mining is unmissable. Among the various mining activities, coal presents a key sector for India, with its labourers being exposed to preventable hazards that result in fatalities and chronic injuries, despite high mechanisation. This paper intends to develop a conceptual framework to theorise workplace safety in Indian coal mining through the lens of preventable accidents and socio-technical alignment by drawing upon the mining accidents database data from the Ministry of Environment, Forest and Climate Change, and literature on occupational safety. The paper is situated within the broader scholarship on occupational safety and highlights how most fatalities occur in ancillary processes such as haulage and transportation, rather than excavation, underscoring that mechanisation alone has not reduced risks. By advancing the concept of *preventable accidents*, the paper emphasises that mining fatalities are not naturalised risks inherent to geology or technology but outcomes of systemic failures in organisational routines, regulatory enforcement, and socio-technical integration. The paper foregrounds the concept of preventable accidents to reframe coal mine hazards not as inevitable but as institutionally produced and therefore avoidable. It also develops a theoretical argument that safety outcomes are shaped less by the degree of mechanisation than by the alignment of socio-technical systems, regulatory regimes, and organisational priorities. Through this framing, the paper situates Indian coal mining within global debates on labour precarity, industrial modernisation, and risk governance. The study challenges fatalistic discourses about mining risk and advances a socio-technical alignment model to guide scholarly debates and policy design. The findings have implications beyond coal mining, offering a framework for understanding safety in other high-risk, labour-intensive sectors where mechanisation coexists with organisational and regulatory deficits.

Keywords Coal mining · Mining labour · Preventable accidents · Socio-technical system

Introduction

With home to over a hundred minerals, India ranks among the top producers of mica, iron ore, bauxite, and coal. Although the mining sector contributes approximately 2.5 per cent of the country's gross domestic product, employing over a million workers (Mukherjee, 2024), mining practices in India have been mired in controversy. According to the Statistical Review of World Energy 2023 (Energy Institute, 2023), India is the second-largest producer, consumer, and coal importer

after China. Nevertheless, it holds the fifth-largest coal reserves, following the USA, Russia, Australia, and China.

Coal mining remains one of the most hazardous industries worldwide, exposing workers to multiple occupational risks, ranging from physical injuries to chronic health conditions. In India, coal mining is critical due to its centrality to the country's energy security, with over 70 per cent of India's electricity generation being coal-based (Ministry of Coal, 2023), and the scale of human labour it employs. Despite advances in mechanisation and regulatory frameworks, coal mining in India continues to witness workplace accidents. According to the Statistical Analysis and Numbers: Key Evaluation of Trends in Coal Mine Accidents (2016–22), there were more than 600 fatalities in the decade between 2013 and 2022. These incidents not only result in the tragic loss of life and livelihood but also stress the

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preventable nature of many mining-related hazards, raising critical questions about organisational accountability, regulatory efficacy, and the socio-economic vulnerabilities of the mining workforce.

The irony of modern Indian coal mining lies in its movement toward mechanisation and its simultaneous dependence on labour-intensive practices (Roy & Schaffartzik, 2021). Although underground mining has progressively transitioned to longwall and continuous mining methods, the dominance of open-cast mining and associated activities continues to expose workers to severe risks¹. Accident records illustrate that fatalities and injuries are disproportionately concentrated in ancillary processes, particularly during transportation and haulage, rather than in core excavation tasks (Bellanca et al., 2021). Such evidence highlights the uneven distribution of risk across mining operations and implies that mechanisation alone cannot guarantee safety if the systemic organisation of work and enforcement of preventive measures remain weak.

The nature of accidents in Indian coal mines must be viewed in a broader political, economic, and institutional context. Historically, coal mining has been a symbol of extractive labour regimes in the Global South, characterised by informalization, inadequate social protection, and insufficient regulatory mechanisms, caught between the imperatives of industrial growth and worker welfare (Lahiri-Dutt, 2018). The preventable nature of most reported accidents reflects systemic deficiencies, such as inadequate training, poor safety culture, ineffective monitoring of safety standards, and insufficient investment in worker-centric technologies. Moreover, the disproportionate vulnerability of contract labourers, mostly migrants, who comprise a substantial share of the mining workforce, adds criticality to the debate on labour precarity and occupational health in India's coal sector (Nayak, 2022).

Theoretically, coal mine accidents can be conceptualised as random events and patterned outcomes of *socio-technical systems* (STS). Introduced by Eric Trist, Ken Bamforth and Fred Emery towards the end of the 1950s to study the coal mining industry in the context of labour studies (Trist, 1981), STS have been defined as the concept “to stress the reciprocal interrelationship between humans and machines and to foster the program of shaping both the technical and the social conditions of work, in such a way that efficiency and humanity

would not contradict each other any longer” (Ropohl, 1999). Over the years, several STS theories have been advanced. Among them, the High Reliability Theory (HRT) and Normal Accident Theory (NAT) offer practical yet contrasting frameworks for explaining how tightly coupled and complex STS, such as those related to mining hazards, function. HRT emphasises that organisations can achieve accident-free operations through rigorous safety culture, training, and redundancy (Weick & Sutcliffe, 2011). In contrast, NAT denotes that accidents are inevitable in complex and tightly coupled systems despite preventive efforts (Perrow, 2010). However, the Indian coal mining experience complicates this dichotomy, as the prevalence of accidents during transportation reflects failures across multiple socio-technical interfaces, including vehicle-operator coordination, signalling systems, traffic management, and maintenance regimes (Dash et al., 2015). Within this framework, transportation-related accidents in India can be understood as outcomes of preventable organisational and regulatory failures, rather than manifestations of inherent system complexity. This positions Indian coal mining as a fertile case for advancing the *theory of preventable accidents* in high-risk industries.

The paper has three objectives. First, it develops a conceptual framework that integrates STS theory, organisational hazard research, and regulatory enforcement literature to explain why many mining accidents are preventable. Second, it analyses descriptive patterns of coal-mining accidents in India to identify the organisational, regulatory, and socio-technical conditions under which recurrent hazards persist. Lastly, building on these theoretical insights and empirical regularities, it proposes a Socio-Technical Safety Alignment Model that explains how misalignments across organisational, technological, demographic, and regulatory domains generate recurring and preventable hazards. Two major contributions are intended through this study. Primarily, the paper foregrounds the conceptual category of *preventable accidents* to examine the persistence of workplace hazards in Indian coal mining. Subsequently, it develops a theoretical argument that safety outcomes in mining are determined less by the level of mechanisation than by the alignment of STS, regulatory enforcement and organisational priorities. This study contributes to wider debates in management, labour studies, and occupational health about the conditions under which industrial accidents become structurally embedded yet potentially avoidable.

Literature review

The study of occupational hazards in coal mining has historically wavered between two perspectives – one that views accidents as inevitable consequences of geological

¹ Risk Assessments, Online Submission & Monitoring of Environmental & CRZ Clearances, Ministry of Environment, Forest and Climate Change Government of India, <https://environmentclearance.nic.in/DownloadPdfFile.aspx?FileName=aXCAorXdOLR7oMCkvQJXYGjrPKdNessRRO5BNurmCIfAcOIhjdZOVCTb8HCivpLH&FilePath=93ZZBm8LWEXfg+HAIQix2fE2t8z/pgnoBhDIYdZCxxVPEh4a7F53Cae7tleKGoXI7f3qHsQ4mEt9+y03oib7i6zg1izg9RsZbVmZBd2bUlrJuQea18z4ICRkFCo2EKv6#:~:text=The%20mining%20activities%20involve%20certain%20types%20of,of%20machinery%2C%20explosion%2C%20to%20name%20a%20few.>

or technological risks, and another that considers them preventable outcomes of organisational and systemic failures. This review engages with both traditions, developing them further to ground the two contributions of this paper. The paper foregrounds the category of *preventable accidents* to reframe mining hazards as avoidable and institutionally produced, arguing that labour safety is determined less by the extent of mechanisation than by the alignment of STS, regulatory enforcement, and organisational priorities.

The concept of preventable accidents

The roots of accident prevention theory can be traced to Heinrich's *Industrial Accident Prevention* (1931), which advanced the notion that most industrial accidents are preventable through eliminating unsafe acts and conditions. Although criticised for its behavioural reductionism, Heinrich's model underlined the preventability of accidents, a theme that subsequent research has developed in more complex ways (Manuele, 2011). Reason's (1990) *Swiss Cheese Model* expanded this logic by demonstrating that accidents occur when multiple system defences fail, thereby emphasising that most accidents are systemic failures rather than random events.

In mining, Hopkins (2005) has shown how catastrophic accidents are often preceded by a series of near misses, which, if addressed, could have averted disaster. These studies collectively argue against the naturalisation of industrial accidents and instead locate them within organisational and institutional domains where prevention is possible. Specifically, in Indian coal mining, many accidents are linked to organisational factors such as inadequate supervision, poor training, and cost-cutting measures. Kunar et al. (2010) further demonstrated that accidents in coal mines are not random but rather result from demographic characteristics of workers, inadequate support systems, and insufficient monitoring. Both studies underscore that fatalities arise from preventable conditions that persist because of weak institutional accountability.

More recent global scholarship has reinforced this perspective. Zhang et al. (2020), analysing Chinese coal mines, identified deficiencies in safety culture as root causes of accidents, emphasising the preventable nature of fatalities even in technologically advanced environments. Similarly, Wu et al. (2023) highlighted the role of weak supervision in China's coal sector, showing that preventable hazards persist when regulatory institutions lack authority or capacity. These findings align with the Indian case, where accidents disproportionately occur during transportation and haulage operations, highlighting subsidiary processes that are highly susceptible to prevention through proper training,

maintenance, and effective regulatory enforcement (Maiti & Khanzode, 2009).

Mechanisation and the persistence of hazards

A vast amount of literature has focused on the role of mechanisation in shaping mining safety. Early socio-technical studies, most notably Trist and Bamforth's (1951) seminal research on the longwall method of coal-getting, revealed that technological innovations radically reconfigured social organisation in mines, often with unintended safety consequences. This insight laid the foundation for STS theory, emphasising that technology and organisation must be optimised for safe and efficient outcomes (Emery & Trist, 1960).

Mechanisation is often assumed to improve safety by reducing direct human exposure to hazardous tasks. For instance, Australian and U.S. mining studies inferred that mechanised systems reduce fatalities linked to manual handling (Burgess-Limerick, 2011; Ruff et al., 2011). In contrast, evidence also shows that mechanisation introduces new risks associated with haulage machinery, conveyor systems, and heavy vehicles (Duarte et al., 2021). In India, haulage accidents remain a leading cause of fatalities, highlighting that mechanisation has changed the characteristics of hazards rather than eliminating them.

Perrow's (2000) theory of *normal accidents* provides a conceptual lens for this paradox. He argued that complex, tightly coupled systems are inherently prone to failure, regardless of technological sophistication. In mining, haulage systems, transportation fleets, and ventilation infrastructures exhibit such coupling, creating vulnerabilities where a single failure can cascade into accidents. Weick and Sutcliffe (2011), through High Reliability Theory, countered that organisations can achieve near-error-free performance in such contexts by institutionalising safety cultures and redundancies. By comparison, empirical studies suggest that neither technological advancements nor safety culture rhetoric have translated into reduced fatalities, precisely because organisational and regulatory misalignments persist (Zhang et al., 2020).

Subcontracting and precarious labour practices further exacerbate this misalignment. Valluru et al. (2017) found that in Australian mines, subcontracted workers experience different and often poorer safety outcomes than permanent employees, primarily due to weaker training, accountability, and oversight. This aligns with Indian evidence showing that a large proportion of fatalities occur among contract workers, who face systemic vulnerabilities (Nayak, 2022). Thus, mechanisation does not automatically translate into improved safety; instead, the alignment of socio-technical, organisational, and regulatory systems determines outcomes.

Regulatory enforcement and institutional weakness

The role of regulation is crucial in preventing preventable accidents. Comparative studies have demonstrated that countries with robust enforcement regimes have lower mining fatalities (Weeks, 1991). In the U.S. and Australia, independent regulators with strong investigative capacities have ensured compliance with safety norms, driving improved labour safety (Gunningham et al., 1998).

In India, regulatory bodies such as the Directorate General of Mines Safety (DGMS) face severe capacity constraints. Sishodiya and Guha (2013) document how enforcement gaps persist due to understaffing, political pressures, and the dominance of production imperatives. This results in a compliance gap, where safety norms exist on paper but are poorly implemented in practice. Empirical research indicates that explosions and gas leaks, although less frequent, are catastrophic when they occur precisely because preventive measures such as ventilation and dust suppression are not adequately enforced (Ray et al., 2022; Shrawankar et al., 2022). These failures reveal that the issue is not one of technological insufficiency but rather institutional weakness. Wu et al. (2023) and Zhang et al. (2020) similarly emphasise that in China, safety improvements stagnate when regulatory enforcement is subordinated to production priorities, which is a dynamic directly mirrored in India.

Methods and data

This study adopts a conceptual research design aimed at theorising workplace safety in Indian coal mining through the lens of preventable accidents and socio-technical alignment. The methodology is grounded in two interrelated assumptions: (i) that workplace accidents are not naturalised risks but socially constructed outcomes that can, in principle, be prevented (Reason, 1990; Hopkins, 2005) and (ii) safety outcomes are mediated less by the degree of mechanisation than by the alignment of demographic, organisational, regulatory, and socio-technical dimensions (Trist & Bamforth, 1951; Perrow, 2000). The study's methodological approach combines descriptive analysis of secondary data with conceptual theorisation to offer a generalisable empirical design.

The Indian mineral accident data from the Environmental Information System (ENVIS) database, which is maintained by the Ministry of Environment, Forests & Climate Change, and the Indian Institute of Technology Dhanbad, are used for the study. The coal mining accident data from 2015 to 2025 are used to understand the causes of accidents at various mining sites. The data from the Ministry of Coal, sourced from the IndiaStat database, is used to map the volume

of coal produced in different states, aiming to understand whether there is a proportional incidence of accidents based on the volume of mining.

The analysis begins with a descriptive mapping of accident categories in the database. The accidents are then categorised into roof and ground collapses, drowning or inundation, haulage-related incidents, machinery failures, explosions, slippage and fall of workers, electric shocks, and medical conditions of the workers. The interpretive categorisation is designed to reveal how accident patterns correspond to systemic misalignments in demographic vulnerabilities, organisational priorities, regulatory enforcement, and socio-technical integration. The paper draws on STS theory and accident prevention literature to conceptually elaborate on how most accidents are, in fact, preventable.

Results

To understand coal mining accidents, data from 2015 to 2025 were analysed for the coal-producing states in India. The causes of fatal accidents are categorised into distinct groups based on the causes noted in the ENVIS database. The distribution of coal-mining-related fatalities across the study period (2015–2025) demonstrates variation in both incidence rates and primary causes, as shown in Fig. 1. The highest overall fatality count occurred in 2015, recording 98 fatalities, dominated by transportation-related causes, which accounted for the maximum over the period ($n=44$). But the trend of fatalities has constantly fluctuated over the years. Overall, the graph shows a substantial decrease in fatalities, with the number rarely exceeding 30 fatalities per year since 2019. The decline after 2015 may be associated with regulatory improvements and technological advancements. Yet, the occasional spikes over the study period highlight that the accidents remained a recurring issue. The low rates of fatalities between 2020 and 2023 may be attributed to the reduced activities during the COVID-19 pandemic. Nevertheless, underreporting during this period cannot be ignored. While the trend over the years shows a reduction in the fatalities, mining in India continues to pose a substantial threat to the workers, and the enduring nature of fatal accidents is a reminder of the fragility of the safety improvements.

While the total annual fatalities showed a general decline post-2015, four major categories persisted as primary concerns – roof/ground collapse, transportation, drowning/flooding, and slippage and fall-related incidents. From Fig. 2, it can be found that more than 60 per cent of the total fatal accidents were caused by roof/ground collapse (35 per cent) and transportation-related causes (26 per cent).

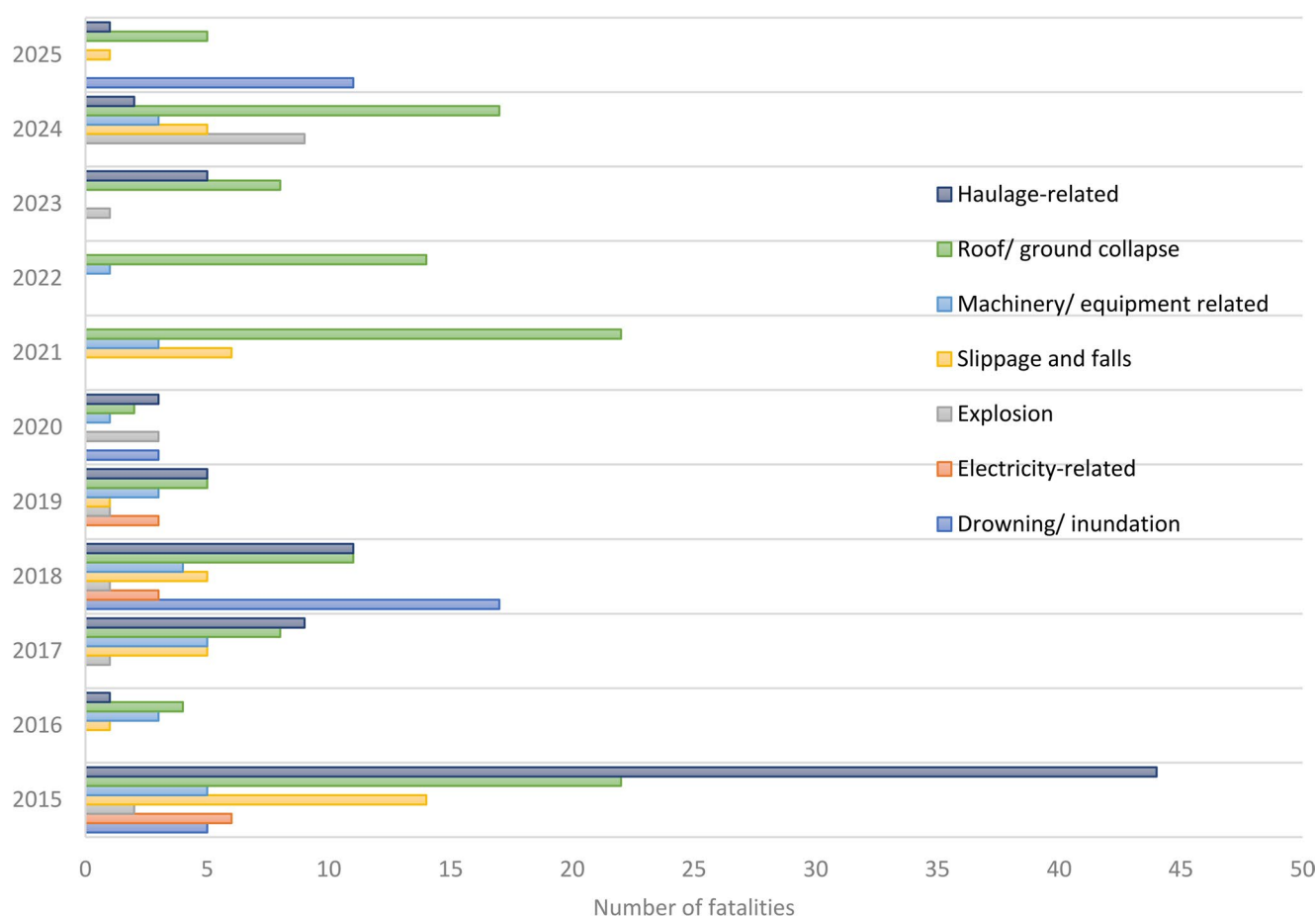


Fig. 1 Number of fatalities in Indian coal mines from 2015 to 2025. Source: Compiled from the ENVIS database

Fatalities resulting from roof and ground collapses underscore the need for enhanced infrastructure, robust support systems, and effective safety monitoring. Although haulage and transportation-related accidents have decreased considerably over the decade, they have persistently contributed to the majority of accidents, emphasising gaps in safety management and vehicle regulations within mines, and highlighting the logistical vulnerabilities in the Indian coal mining industry.

Fatalities due to machinery-related accidents, explosions, and other causes are consistently lower in proportion throughout the decade, suggesting that risk mitigation strategies should be primarily focused on the four highest-frequency hazards, which together contribute to 84 per cent of the total number of fatalities. The risks in workers' lives due to falls and equipment incidents further demonstrate the shortcomings in workplace safety procedures and training for workers. Although catastrophic incidents attract public attention, most mining fatalities are caused by routine operating hazards rather than infrequent disasters, as seen by the lower incidence of blasting or electricity-related accidents. This trend supports the idea that negligence and poor

enforcement of safety measures are more systemic problems than isolated incidents regarding mining safety in India. In essence, the uneven distribution of mining hazards across the country emphasises the critical need for state-specific safety measures.

The distribution of mining fatalities across the top ten coal-producing states in India – Odisha, Chhattisgarh, Jharkhand, Madhya Pradesh, Telangana, Maharashtra, West Bengal, Uttar Pradesh, Meghalaya, and Assam – between 2015 and 2025 is depicted in Fig. 3. Further comparison of the data shows that, except for Madhya Pradesh, Odisha, Chhattisgarh and Maharashtra, all the other states have a higher incidence of fatalities per million tonnes of coal produced (Fig. 4). The higher number of fatalities points toward either an increase in mining intensity or possible deficiencies in operational safety enforcement. Regional variations in accident frequency are further highlighted by the higher number of fatalities in the north-eastern states, such as Assam and Meghalaya, owing to their terrain. Even though the National Green Tribunal in India had banned illegal mining activities in the north-eastern states since 2014, the incidence of accidents has been high in the area,

Fig. 2 Categorisation of coal mine accidents over the decade.
Source: Compiled from the ENVIS database

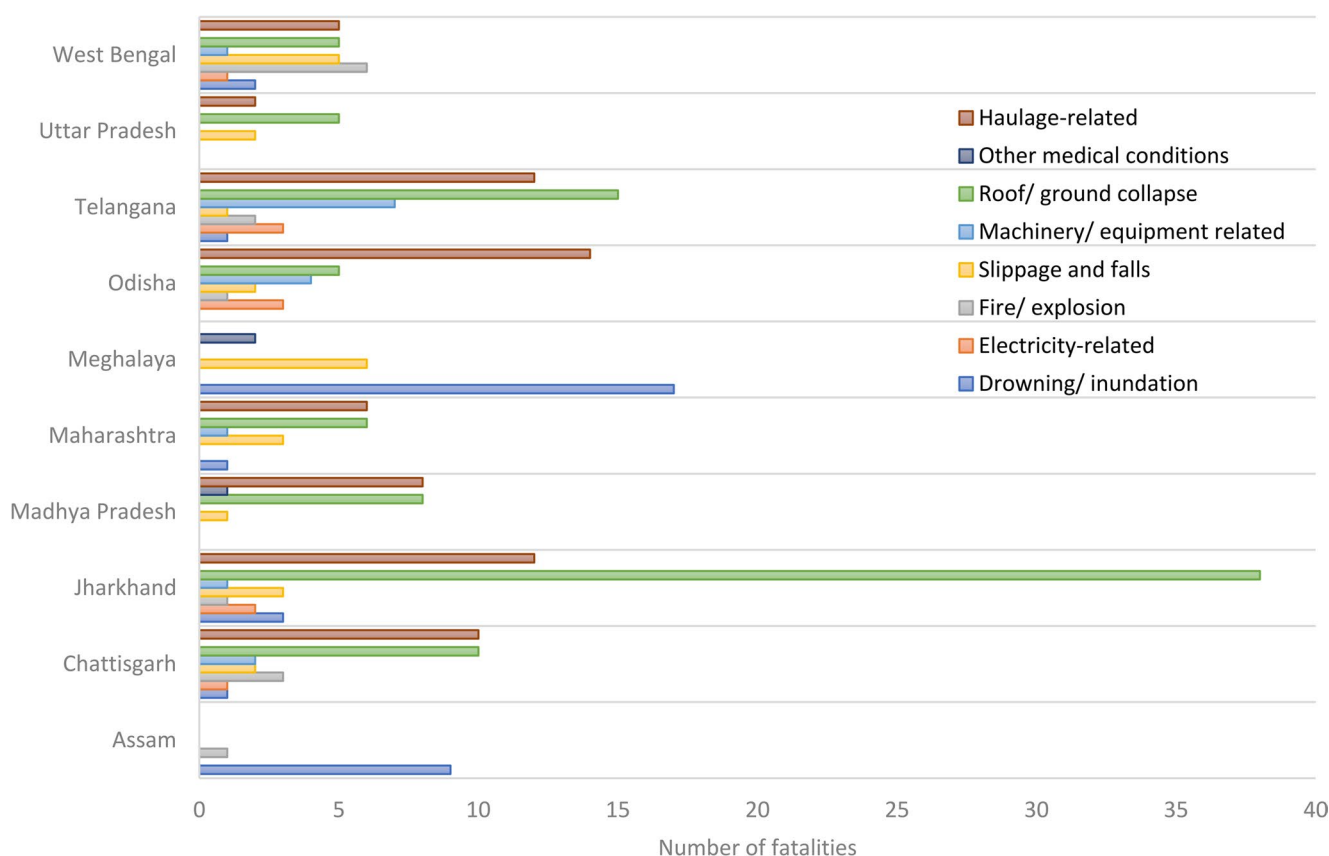
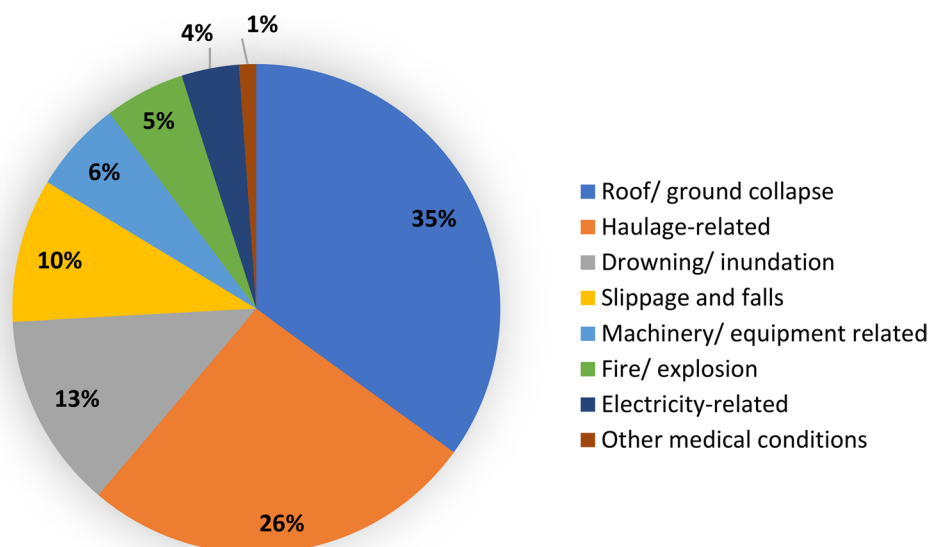


Fig. 3 Type of fatalities across states from 2015 to 2025. Source: Compiled from the ENVIS database

mainly due to underlying flaws in labour safety and regulations being implemented in the hilly terrains, where rat-hole mining issues have been widely reported.

Data shows that India has made progress in reducing mining accidents since 2015, reflecting improved regulations and enhanced safety enforcement. Nonetheless, critical

safety concerns persist. The varying patterns imply that regulatory benefits are not consistently maintained across geographies or types of mining. While the rates have decreased over time, roof and ground collapses remain a serious threat, highlighting deficiencies in structural evaluations, preventive care, and the application of safety support systems. The

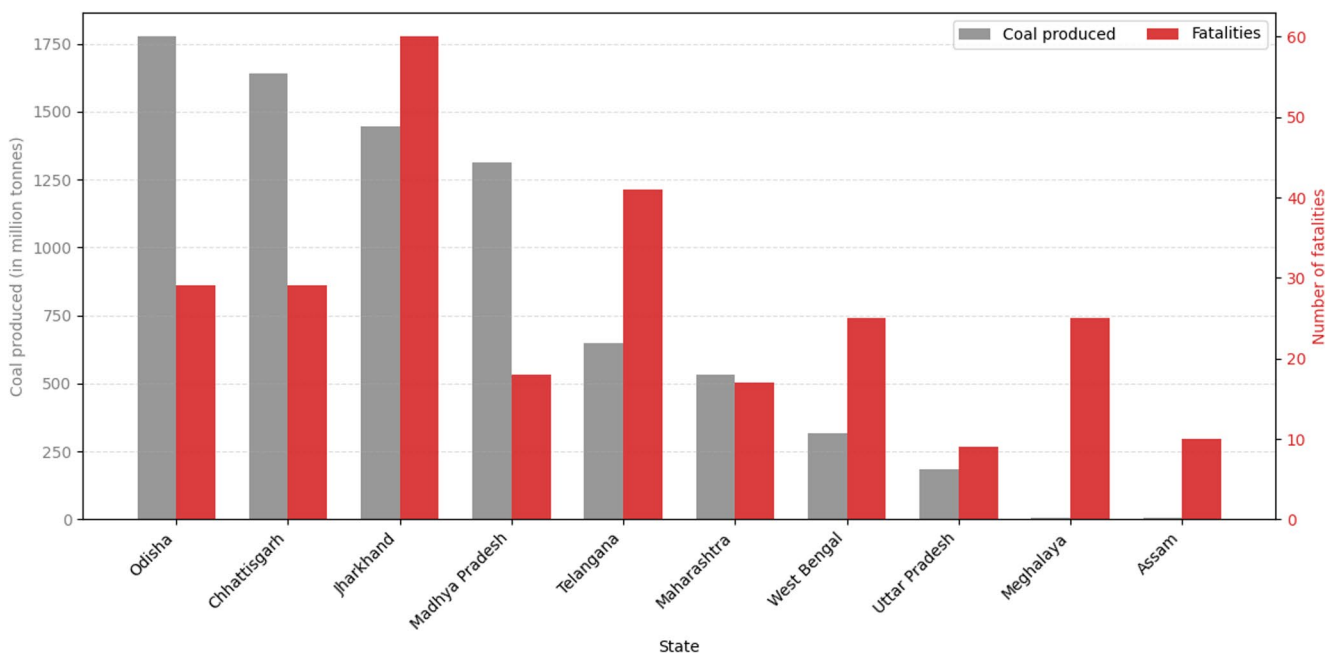


Fig. 4 Coal production and fatal accidents across states from 2015 to 2025. Source: Compiled from the ENVIS database and the Ministry of Coal, Government of India data

frequency of haulage-related accidents remains a leading cause of fatality over time. Vehicle accidents in a controlled mine environment should be preventable, unlike collapses, which geological and seasonal factors, such as monsoons can cause. The persistent occurrence of these incidents shows that mine traffic regulations are a shortcoming and that safety compliance is not adequately monitored (Dash et al., 2015). Deaths related to falls, handling equipment, and working during unsupervised hours (such as after-evening shifts) expose a lack of emergency response facilities, safety nets, and worker training (Hopkins, 2005). Rather than resulting from inherent mining risks, these deaths are preventable and reflect systemic negligence. Differences at the state level underline that mining intensity alone does not determine safety. The unusually high death toll in some states points to differences in labour conditions, contractor practices, and safety code enforcement. Overall, the results indicate that routine and avoidable operational hazards continue to be a leading cause of death in Indian mines, even though catastrophic events such as fires and explosions are comparatively uncommon, given that they have not been reported.

Discussion

The discussion interprets the descriptive patterns through the study's central argument that coal-mining accidents in India are preventable and arise from systemic misalignment

across socio-technical, organisational, demographic, and regulatory domains. The results demonstrate that the highest-frequency hazards – roof collapses, haulage failures, falls, and flooding – align directly with known fault points in socio-technical integration and enforcement regimes, supporting the premise that fatalities are institutionally produced rather than inherent to the mining industry. The persistence of fatalities suggests that accidents are not inherent to mining operations, but instead institutionally produced when organisational priorities, labour arrangements, and enforcement mechanisms fail to align with the technological and ecological demands of the mining system (Newnam & Dee, 2026). Situating these findings within international evidence further reveals that India's accident patterns align with global regularities in high-risk sectors, where mechanisation advances more rapidly than safety governance. This provides the conceptual grounding for the proposed Socio-Technical Safety Alignment Model (ST-SAM), which explains why preventable hazards persist despite technological progress.

The patterns observed in Indian coal mining closely mirror international evidence. For instance, the dominance of haulage accidents parallels findings from the United States of America, where mechanisation has shifted risks toward transportation systems rather than eliminated them (Bellanca et al., 2021; Ruff et al., 2011). Similarly, the recurrence of roof and ground collapses is consistent with safety failures documented in China and South Africa, where structural support and enforcement gaps remain key contributors

to fatalities (Zhang et al., 2020; Pelders & Nelson, 2019). State-level variation in India resembles regional disparities in South Africa and China, where weaker regulatory oversight corresponds to higher fatality rates. These parallels position India's coal-mining hazards within established global patterns of socio-technical misalignment in high-risk industries.

Socio-technical safety alignment to Indian coal mine accidents: A theoretical framework

The high levels of fatalities in Indian coal mining, in spite of decades of mechanisation and regulatory codification, demand a theoretical lens that moves beyond technological determinism. As mentioned, the paper proposes an ST-SAM that conceptualises mining accidents as preventable outcomes of systemic misalignments across four critical dimensions: demographic vulnerabilities, organisational priorities, regulatory enforcement, and STS (Fig. 5). This framework reframes accidents not as stochastic or inevitable but as socially produced alignment failures. To ground this framework empirically, the theoretical dimensions are mapped onto the major categories of accidents, including roof falls, haulage accidents, machinery failures, explosions, and transportation-related incidents.

By situating accident categories within the ST-SAM, the model not only integrates the empirical realities of Indian coal mining but also demonstrates how each accident category demonstrates systemic faults. The study highlights

how safety outcomes are mediated by factors such as labour composition, organisational imperatives, enforcement weaknesses, and the socio-technical embedding of mechanisation and also shows how each accident category represents preventable accidents.

Demographic vulnerabilities and regulatory gaps

Many underground workers are contract labourers recruited from marginalised communities with minimal formal training. As seen in the case of mine workers in South Africa, the limited literacy and technical knowledge of the labourers impede the correct application of roof support systems, such as bolting or props, especially in highly mechanised mines where monitoring systems require careful calibration (Pelders & Nelson, 2019). Research in Chinese coal mines has shown that unforeseen natural hazards rarely cause accidents involving roof falls, haulage, and transportation; instead, they are often attributed to inadequate support systems, poor geological assessments, and failures in supervisory oversight (Zhang et al., 2020). These failures are preventable if regulatory bodies strictly enforce roof support standards and companies invest systematically in worker training, as was evident in countries like the USA and China (Wu et al., 2023; Saleh & Cummings, 2011). These vulnerabilities intersect with organisational priorities that shape how safety is managed on the ground. The continuance of roof falls, even when their causes are known and preventive technologies are available, underscores the

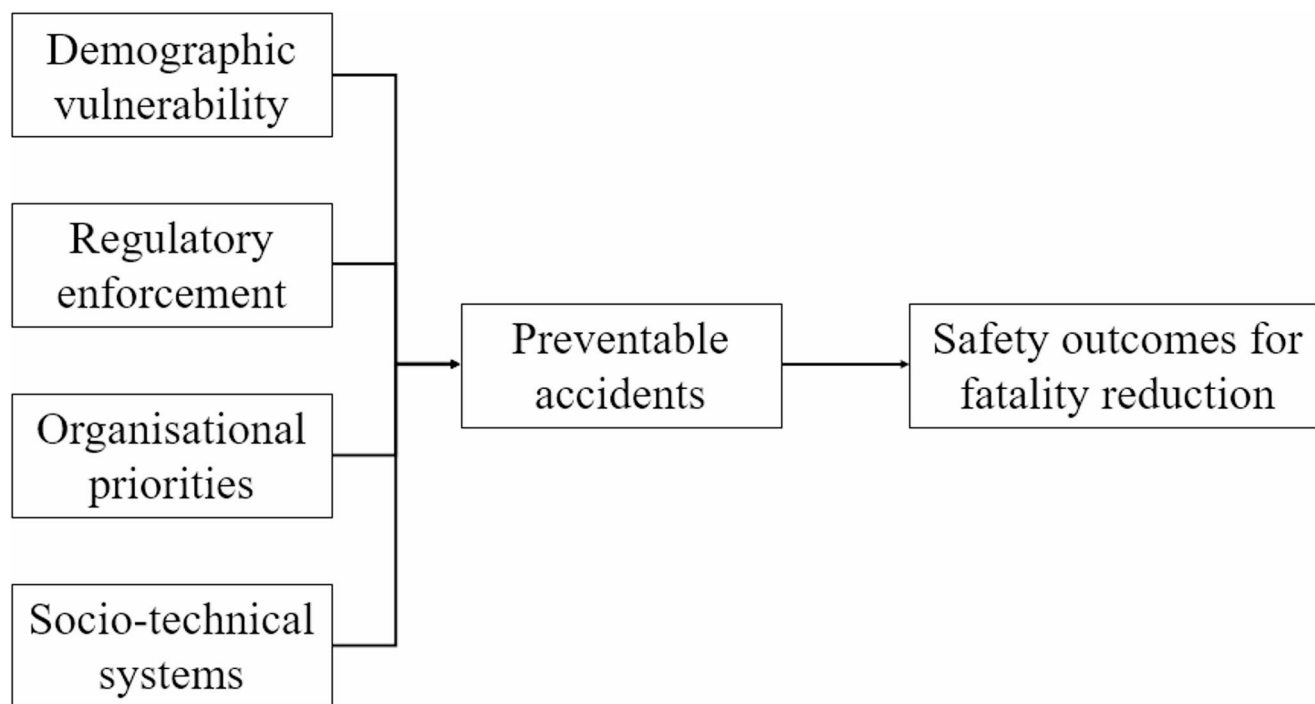


Fig. 5 Socio-Technical Safety Alignment Model (ST-SAM) for coal mine safety. Source: Conceptual model constructed for this study

importance of reclassifying them as preventable accidents. This reclassification highlights institutional responsibility rather than fatalistic acceptance of hazards by acknowledging the vulnerability of the mining labour force.

Organisational priorities and socio-technical misalignment

Coal mine accidents are deeply entwined with organisational priorities and socio-technical misalignments. Mechanisation has intensified haulage operations, with conveyor belts, locomotives, and heavy earth-moving machinery increasingly central to coal extraction. However, the integration of such machinery into organisational routines has been fraught with misalignment. Thompson et al. (1998) and Pandey and Mishra (2023) present cases from the contexts of South Africa and India, respectively, illustrating that haulage accidents often result from inadequate signalling systems, poor traffic management, and operator fatigue. Organisational pressures to meet production targets exacerbate these risks, as operators are implicitly or explicitly encouraged to work long hours with minimal rest (Erkan et al., 2016). Moreover, Australian cases demonstrate that subcontracting practices shift responsibility for maintenance and safety inspections of haulage systems, as subcontracted workers often operate machinery with insufficient training, thereby increasing the likelihood of accidents (Valluru et al., 2017, 2020).

Theoretically, haulage accidents exemplify how mechanisation does not inherently improve safety but can generate new vulnerabilities if not aligned with organisational and regulatory systems. Perrow's (2000) theory of everyday accidents is instructive because tightly coupled systems, such as haulage machinery in mines, create inevitable failures unless safety is systematically prioritised. The Indian case shows that organisational imperatives, particularly the prioritisation of production over safety, prevent this alignment and perpetuate preventable fatalities. Thus, it can be seen that organisational gaps in maintenance and monitoring directly manifest as socio-technical failures in daily operations.

Socio-technical integration and maintenance deficits

Machinery-related accidents, such as equipment malfunctions or operator entanglements, highlight the challenges of socio-technical integration. In highly mechanised mines, machinery failures represent one of the most visible categories of preventable accidents. The data indicate that while mechanisation has reduced some traditional risks, it has simultaneously introduced machinery-specific vulnerabilities. Accidents caused by machinery are often preventable through regular maintenance, adequate training,

and the consistent use of protective equipment. Nevertheless, organisational cost-cutting and inadequate regulatory enforcement compromise these safeguards. Studying machine-related accidents in Canada, Chinniah (2015) mentions that behavioural factors, such as operators bypassing safety procedures to accelerate production, significantly contribute to machinery accidents. These behavioural factors are not individual shortcomings but organisationally induced patterns, shaped by cultures that privilege speed and output over compliance with safety norms.

From a socio-technical perspective, machinery failures are emblematic of misalignment between technology and the social systems intended to manage it. Beck's (1992) notion of *manufactured risks* applies directly here, that modernisation introduces new risks that are neither natural nor inevitable but socially produced by adopting technology without corresponding safety infrastructures.

Regulatory enforcement and institutional failures

Explosions in coal mines, whether caused by gas leaks, blasting errors, or the accumulation of coal dust, continue to be a major source of catastrophic accidents. Even though less frequent than roof falls or haulage incidents, explosions cause disproportionately high fatalities when they occur. Explosions highlight the role of regulatory enforcement in shaping safety outcomes. Stringent safety standards are in place for ventilation, blasting protocols, and dust suppression. Nonetheless, their implementation is often negligent. Sahu & Mishra (2023) observe that most explosions in Indian mines are preventable with existing technologies and procedures, but persist due to weak enforcement and inadequate inspections. In many cases, regulatory agencies lack the necessary personnel and resources to ensure compliance, and political and economic pressures discourage stringent oversight of coal companies tasked with meeting national energy demands. Preventable mine accidents are not technological inevitabilities but institutional failures. They reveal how regulatory enforcement, when subordinated to production imperatives and undermined by compliance gaps, allows avoidable threats to escalate into disasters. Comparative research by Wildfire (2012) in the USA reinforces this point, showing that fatalities can decline in jurisdictions with robust enforcement and independent oversight.

Mining accidents as intersections of demography, organisation, and technology

Transportation accidents, particularly those involving surface movement of coal and workers, constitute the primary category of fatalities in the dataset used for the study. These incidents often occur during the movement of heavy

vehicles within mines or coal transport to external facilities. Workers from marginalised groups, frequently employed as drivers or support staff, face precarious working conditions with limited training. Organisational cost-cutting leads to poor vehicle maintenance and inadequate traffic regulation inside mining sites. Regulatory oversight of surface transportation is often weaker than that of underground activities, further compounding risks. Studying Australian mines, Laurence (2005) notes that many accidents occur during shift changes, when a large number of workers are moved simultaneously, overwhelming existing safety protocols and creating communication gaps.

Proper maintenance regimes, traffic management systems, and rigorous training could drastically reduce these accidents. Their recurrence underscores the systemic misalignment of social, organisational, and technical factors that ST-SAM seeks to explain. The mapping confirms that fatalities across these categories are not random occurrences but systematically preventable accidents. Each category, when understood through the ST-SAM, demonstrates how safety outcomes depend less on the degree of mechanisation and more on the alignment of demographic, organisational, regulatory, and socio-technical dimensions. Thus, these insights collectively demonstrate the explanatory value of ST-SAM.

Conclusion

The study argues that Indian coal mining accidents, while often normalised as inevitable, are overwhelmingly preventable. By developing the concept of preventable accidents, the paper reframes mining fatalities as socially produced outcomes of organisational negligence, weak enforcement, and socio-technical misalignments rather than inevitable risks. The ST-SAM demonstrates that labour safety depends not merely on mechanisation but on integrating demographic realities, organisational priorities, and regulatory enforcement. Theoretically, the paper contributes to occupational safety scholarship by advancing a middle position between HRT and NAT, showing that while complex systems entail risks, institutional failures in even routine subsystems perpetuate hazards. Practically, it advocates for a shift from technological determinism to systemic reforms that prioritise safety culture, regulatory independence, and worker empowerment.

Theoretical and practical implications

This paper contributes to STS research by clarifying the conditions under which accidents remain preventable despite mechanisation. The evidence shows that neither

HRT nor NAT fully explains India's accident patterns. Hazards persist not because systems are inherently complex (as NAT would predict), nor because safety culture and redundancy are fully implemented (as HRT requires), but because systemic misalignment across organisational, regulatory, demographic, and technological domains undermines risk controls. The proposed ST-SAM advances STS theory by demonstrating that preventable accidents emerge when organisational priorities and regulatory enforcement fail to keep pace with technological change. This position prevents the preventability not as an operational issue alone but as a socio-institutional outcome, extending global debates on risk governance (Perrow, 2000; Weick & Sutcliffe, 2011).

The findings carry several practical implications for improving safety governance in Indian coal mining. The frequent roof and ground collapses emphasise the need for targeted inspections, stricter compliance with support-system standards, and continuous training for contract workers, whose vulnerabilities mirror patterns observed in South Africa and China. The predominance of haulage-related fatalities indicates that mine management should strengthen traffic control systems, vehicle maintenance regimes, and operator fatigue monitoring, aligning with international best practices documented in countries such as China and the US. The state-level variation in fatality rates underscores the pressing need for capacity-building in regulatory bodies, such as the DGMS, enhanced contractor oversight, and transparent reporting systems that minimise underreporting. Essentially, for policymakers, the findings emphasise the urgency of strengthening regulatory capacity, enforcing transparency in accident reporting, and embedding safety within organisational incentives. Only through such alignment can India's mining sector reconcile its role in national energy security with the dignity, safety, and well-being of its labour force.

Limitations and future research

By grounding accident causation in institutional and socio-technical misalignment, the study advances a framework that can guide both research and regulatory reform in high-risk sectors. Acknowledging the limitations of the available data, future research can build on this study by adopting mixed-methods approaches that complement descriptive accident records with qualitative insights. Ethnographic fieldwork, worker interviews, and organisational audits would deepen understanding of how miners experience risk, how supervisors operationalise safety routines, and how regulatory gaps manifest in daily work practices. Comparative studies across jurisdictions with stronger enforcement regimes, such as Australia, the United States, and South Africa, would further clarify how institutional

capacity, inspection quality, and supervisory consistency influence preventable accidents. Examining the role of emerging technologies, including AI-driven predictive analytics, IoT-based hazard detection, and automated haulage systems, could also illuminate how digitalisation interacts with socio-technical alignment. Finally, future research should incorporate a gender and inclusion lens, recognising how gender-differentiated labour policies and the experiences of women in mining reshape vulnerability and safety outcomes. These directions would not only refine ST-SAM but also help develop a more comprehensive and inclusive understanding of mining safety in India.

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Data Availability http://www.ismenvis.nic.in/Database/Mining_Accidents_in_India_24483.aspx<https://www.indiastat.com/><https://www.coal.nic.in>

Declarations

Competing interests The authors declare no competing interests.

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